

Experiment 7

7.1.2 Possible Combinations from Three Digits

Algorithm :

Step 1: Start

Step 2: Input a, b, c

Step 3: Create list to store digits

digits = [a,b,c]

Step 4 : Set loop a in range 0 to 3

Step 5 : Set loop b in range 0 to 3

Step 6 : Set loop c in range 0 to 3

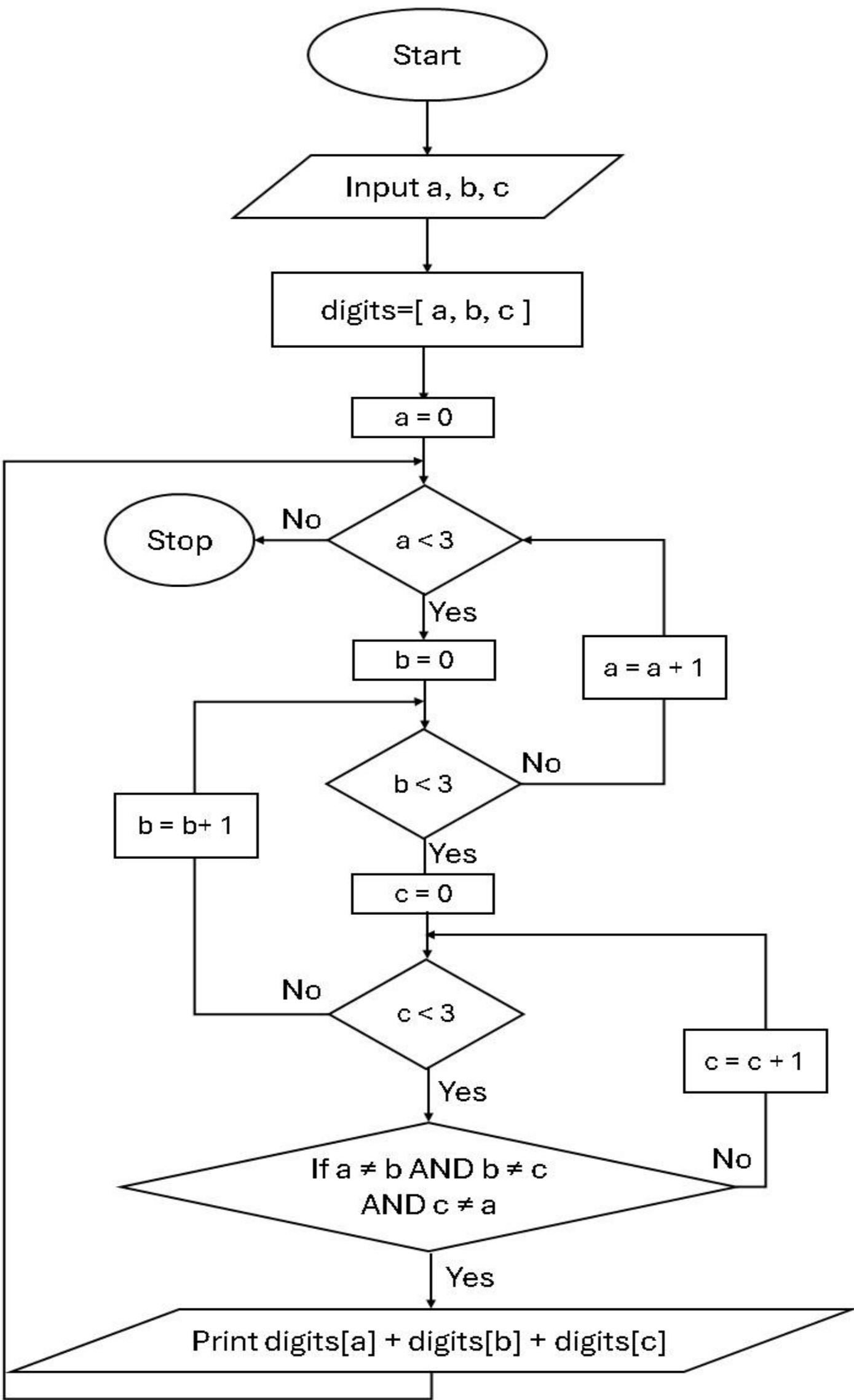
Step 7 : if $a \neq b$ AND $b \neq c$ AND $c \neq a$

then Print digits[a] + digits[b] + digits[c]

Step 8 : Repeat Step 4, 5, 6

Step 9 : Stop

Flowchart :



```
a,b,c=input().split()
digits=[a,b,c]
for a in range(3):
    for b in range(3):
        for c in range(3):
            if a!=b and b!=c and c!=a:
                print(digits[a]+digits[b]+digits[c])
```

Execution :

CODETANTRA

Home

7.1.2. Possible Combinations from Three Digits

Write a Python program to accept three digits and print all possible 3-digit combinations that can be formed using those three digits. Each digit should be used exactly once in each combination.

For three digits $a, b,$ and $c,$ there are 6 possible arrangements based on their indices: $abc, acb, bac, bca, cab, cba.$

Input Format:

- Single line contains three space-separated non-negative integers, $a, b,$ and $c,$ representing the three digits

Output Format:

- Print all possible 3-digit combinations
- Each combination should be printed as a 3-digit number on a separate line
- Print combinations as they appear (maintain leading zeros if first digit is 0)

Sample Test Cases

28:53

🔍 🌙 📄

combinati...

1

a,b,c=input().split()

2

digits=[a,b,c]

3

for a in range(3):

4

for b in range(3):

5

for c in range(3):

6

if a!=b and b!=c and c!=a:

7

print(digits[a]+digits[b]+digits[c])

Average time

0.004 s

Maximum time

0.006 s

3 out of 3 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1

6 ms

Test case 2

4 ms

Test case 3

4 ms

Terminal

Test cases

< Prev

Reset

Submit

Next >